1. LLM Intro
2. AI(ML(DL(LLM)))
3. Intro to Pretraining & finetuning
4. Transformer (BERT v/s GPT)
5. Overview of LLM
6. Open AI's model's property
7. zero shot v/s few shot learning
8. Open source v/s closed source
9. Unsupervised v/s self supervised
10. Autoregressive Model
11. Emergent Behaviour
12. STAGES OF BUILDING LLM ✓
13. Summary.

* Data preparation & Sampling.
1. Tokenisation.
   i) Downloading/Implementing Data
   ii) Token → Token ID.
      · Tokenizing & de-tokenizing
   iii) Special Context token <|unk|> <|endoftext|>
2. Byte Pair Encoding.
   i) Sub-word based tokenizer
3. TikToken
4. Input - Target pair
5. Dataset & Dataloader
   i) Stride, slide, Context_len, max_len, etc.
6. Vector Embedding
   i) Embedding Matrix
   ii) Embedding V/S Linear layer
7. Positional Embedding
   Absolute V/S Relative Posit. Embed.
8. Input Embedding Vector.
Data preparation & Sampling done ✓

* Attention Mechanism.
1) Types of Attention
2. RNN and its limitations.
3. Bahdanau's attention
4. Context vector
5. Self-Attentn Mechanism
   i) without trainable weight
   ii) With trainable weights
6. Normalization & Softmax
7. Attention Scores & Weights
8. Trainable weights W_q, W_k, W_v
9. Attention Score & weights
10. Scaling & softmax.
11. Context vector Matrix.
12. Diagram.
13. Casual Attention.
14. Dropout
15. Multihead Attention
   i) with & without Weight & Split
Attention Complete ✓
Understand dimensions.

* LLM Architecture.
1. Overview
2. Layer Normalization.
3. GELU
4. Linear layer
5. Feed forward
6. Short circuit
7. Vanishing gradient prob
8. Predicting next word.
9. WorkFlow
LLM Building done ✓

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* Training
1) * Training loss
1) Cross Entropy loss
2) Perplexity
3) Training & Validation losses
4) Working.
2. LLM pretraining LOOP.
* 1) Parameters in GPT
2) Updating parameters
3. Decoding Techniques
1) Temperature Scalling
2) Top-K

* Saving & loading Weights

* Pretrained weights by OpenAI

* Finetuning
1) Types of finetuning.